Class #12 - Orthogonal Projections

1. An $m \times n$ matrix U has orthonormal columns if and only if $U^TU = I$.

   Proof:



Want:

$\|u_i\|^2 = 1 \quad \forall i$

$u_i \cdot u_j = 0 \quad \forall i \neq j$

$$u_i \cdot u_j = \begin{cases} 1 & i=j \\ 0 & \text{otherwise} \end{cases} \quad (*)$$

$(\Rightarrow)$ $U \in \mathbb{R}^{m\times n}$ $\{u_1, \dots, u_n\}$ is an orthonormal set.

To show: $U^TU = I$

$$U^TU = \begin{bmatrix} - u_1^T - \\ - u_2^T - \\ \vdots \\ - u_n^T - \end{bmatrix} \begin{bmatrix} | & & | \\ u_1 & \cdots & u_n \\ | & & | \end{bmatrix} = \begin{bmatrix} u_1^Tu_1 & u_1^Tu_2 & \cdots \\ & \ddots & \\ & & \end{bmatrix} = I$$

Orthogonal matrix

$(*)$ $u_i \cdot u_j = \begin{cases} 1 & i=j \\ 0 & i\neq j \end{cases}$

$m \times n$ orthonormal columns $\Rightarrow U^TU = I$ ($n\times m \cdot m\times n$)

$\Leftrightarrow U^TU = I$

$(\Leftarrow)$ Given: $U \in \mathbb{R}^{m\times n}$, $U^TU = I$

$[U^TU]_{ij} = \begin{cases} 1 & i=j \\ 0 & i\neq j \end{cases}$

$[U^TU]_{ij} = u_i \cdot u_j$

$\therefore \{u_1, \dots, u_n\}$ is an orthonormal set

$\therefore U$ has orthonormal columns.

2. Let U be an $m \times n$ matrix with orthonormal columns, and let x and y be in $\mathbf{R^n}$. Then,

   (a) $||Ux|| =$ $\|x\|$

   (b) $(Ux) \cdot (Uy) =$ $x \cdot y$

   (c) $(Ux) \cdot (Uy) = 0$ if and only if $x \cdot y = 0$ ($\Leftrightarrow$ x and y orthogonal)

   Proof:

$\|Ux\|^2 = (Ux)\cdot(Ux) = (Ux)^T(Ux) = x^T(U^TU)x = x^TIx = x\cdot x = \|x\|^2$

3. An *orthogonal matrix* is a square _____ matrix U such that $U^{-1} =$ _____.
Fill in the blank: An orthogonal matrix has _____ columns.
Fill in the blank: Any square matrix with orthonormal columns is an orthogonal matrix, and such a matrix has _____ rows!

4. Example: Let $\{u_1, u_2, \cdots, u_5\}$ be an orthogonal basis for $\mathbf{R}^5$ and let

$$y = c_1u_1 + c_2u_2 + \cdots + c_5u_5.$$

Consider the subspace $W = \text{Span}\{u_1, u_2\}$, and write y as the sum of a vector z_1 in W and a vector z_2 in $W^\perp$.

5. The Orthogonal Decomposition Theorem: Let W be a subspace of $\mathbf{R}^n$. Then each y in $\mathbf{R}^n$ can be written uniquely in the form

$$y = \hat{y} + z$$

where _____ is in _____ and _____ is in _____. In fact, if $\{u_1, \cdots, u_p\}$ is any orthogonal basis of W, then

$$\hat{y} = _____$$

and $z = y - \hat{y}$.

6. Fill in the blank: If y is in $W = \text{Span}\{u_1, \cdots, u_p\}$, then orthogonal projection of y onto $W =$ _____.

$$\hat{y} = \left(\frac{y \cdot u_1}{u_1 \cdot u_1}\right) u_1 + \cdots + \left(\frac{y \cdot u_p}{u_p \cdot u_p}\right) u_p$$

$y = \hat{y} + z$, $\hat{y} \in W$, $z \in W^{\perp}$. If $y \in W$, then $\hat{y} = y$.

7. The Best Approximation Theorem: Let W be a subspace of $\mathbf{R^n}$, y any vector in $\mathbf{R^n}$, and $\hat{y}$ the orthogonal projection of y onto W. Then _____ is the closest point in W to y, in the sense that

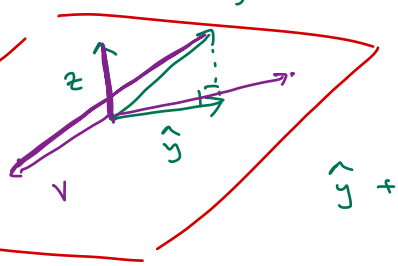
$$\|y - \hat{y}\| < \|y - v\|$$

$\hat{y}$: closest vector to y in W

for all v in W distinct from _____.

Proof:

Will show: $\|y - v\|^2 - \|y - \hat{y}\|^2 > 0$



$\hat{y} + z = y$

$$\text{LHS} = \|(\hat{y} + z) - v\|^2 - \|y - \hat{y}\|^2$$

$$= \|(\hat{y} - v) + (z)\|^2 - \|z\|^2$$

($\hat{y} - v \in W$, $z \in W^{\perp}$)

$$= \text{(Pythagorean)} \quad \|\hat{y} - v\|^2 + \|z\|^2 - \|z\|^2 = \|\hat{y} - v\|^2 > 0$$

$(v \neq \hat{y})$

8. Example: If $u_1 = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}$, $u_2 = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$, $y = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, and $W = \text{Span}\{u_1, u_2\}$, then the closest point in W to y is:

$\{u_1, u_2\}$ orthogonal basis of W

$\hat{y}$ (orthogonal proj of y onto W)

$$= \left(\frac{y \cdot u_1}{u_1 \cdot u_1}\right) u_1 + \left(\frac{y \cdot u_2}{u_2 \cdot u_2}\right) u_2$$

Can be calculated.

9. The distance from a point y in $\mathbf{R^n}$ to a subspace W is defined as the distance from y to the nearest point in W. Find the distance from y to W above.

$= \|z\| = \|y - \hat{y}\|$

$\hat{y}$ (orthogonal proj of y onto W)

Distance of y to the subspace W

$= \|z\| = \|y - \hat{y}\|$

① Attendance

② Par. quiz

③ Class # 12 #7 ~

④ will discuss Midtem. Highest: 100
Avg. : 72.05

⑤ Q#6 (Nov. 9)

① Pythagorean Theorem:

Two vectors u, v are [orthogonal]

$\Rightarrow$
$\Leftarrow$
$\Leftrightarrow$

$$\|u+v\|^2 = \|u\|^2 + \|v\|^2$$

② Orthogonal Decomposition Theorem:

W : subspace of $\mathbb{R}^n$

Then, each y in $\mathbb{R}^n$ can be written uniquely

$$y = \hat{y} + z$$

where $\hat{y} \in W$

$z \in W^\perp$

If $\{u_1, \dots, u_p\}$ orthogonal basis of W,

then $\hat{y} = \boxed{\frac{y \cdot u_1}{u_1 \cdot u_1} u_1 + \cdots + \frac{y \cdot u_p}{u_p \cdot u_p} u_p}$

Name: orthogonal projection of y onto W

and $z = \boxed{y - \hat{y}}$

In particular, if $y \in W$, then $\hat{y} = y$ and $z = \boxed{0}$